

Instructor & Textbook Information

Welcome

Advancements in hardware and software have made data easily available for use. This introductory analytics course starts with an overview of data analytics to introduce the different types of data and fundamental concepts behind collecting, storing, and analyzing data. A survey of the different types of analytics (descriptive, diagnostic, predictive, and prescriptive) will provide insight into how each type can be applied to derive the most benefit from available data. Use cases will be discussed to evaluate best practices and lessons learned.

Instructor Information

Name: Yu Zhang

Email: Use Canvas inbox to email instructor

Yu Zhang has over 5 years of experience in python programming and uses the big query SQL, python colab, and Tableau in previous and current work. She has over 5 years of experience in academia in statistical analysis and 3 years of experience in industry and government working on complex data and machine learning projects. She is technically sound with full experience in data validation and machine learning modeling. In addition, she has over 6 year's teaching experience as a lecturer, teaching assistant, and mentor working in UC Santa Barbara, UC Davis, UC Irvine. She is passionate about using data analytics for real-world problem solving and working with diverse students.

Course Material

This course requires Microsoft Excel. Students who do not have access to Microsoft Excel may activate a free 30-day trial and purchase the Microsoft Office Suite from any retailer of their choice, we recommend the UCI Textbook store: <https://uci.bncollege.com/shop/uci/home>. 
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There are no required materials for this course. Below are **optional** but recommended textbooks.

- Delen, D. (2021). *Predictive Analytics: Data Mining, Machine Learning and Data Science for Practitioners 2nd Edition*. Upper Saddle River, New Jersey: Pearson (FT Press Analytics)
- Delen, D. (2019). *Prescriptive Analytics: The Final Frontier for Evidence-Based Management and Optimal Decision Making*. FT Press.

Course Etiquette

Be Prepared: Make sure to complete all required work prior to attending scheduled meetings.

Be Attentive: Be present and attentive throughout the class. Engage in discussion, participate in activities, and ask questions. In virtual sessions, find a quiet location so unwanted background noise and other distractions are at a minimum.

Be Respectful: When interacting with classmates and instructors, consider your communication style; including, but not limited to, phrasing, tone, and text formatting. Also, be respectful of different ideas and opinions.

[Click Next \(below\) to progress through the course.](#)